

Recommendations — mcrc-kras-a59t-pik3ca-peritoneal-pre-hi pec-q8k4

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

Generated 2026-05-29

Recommendations — mcrc-kras-a59t-pik3ca-peritoneal-pre-hipec

Profile snapshot

- **Primary site:** colon
- **Histology:** colon adenocarcinoma (presumed; histologic subtype not specified at intake)
- **Stage:** IV (metastatic; liver metastases status post-resection, peritoneal involvement present)
- **ECOG:** 1
- **Age band:** 30-39
- **Sex:** unknown
- **Biomarkers:** Germline hereditary cancer panel negative; KRAS A59T mutated (atypical, switch-II region; non-G12/G13/Q61) (ngs_pending); APC E1295 mutated (likely truncating; presumed mutation cluster region) (ngs_pending); TP53 R273 mutated (hotspot; specific substitution not specified) (ngs_pending); PIK3CA M1043I mutated (kinase-domain activating) (ngs_pending); SMAD4 R361H mutated (MH2-domain hotspot; loss-of-function) (ngs_pending); 20q11 amplification (BCL2L1/TOP1 co-amplified) amplified (ngs_pending); MSI status MSS; TMB low; PD-L1 negative; HER2 (ERBB2) amplification not reported (ngs_pending); BRAF V600E not explicitly reported (ngs_pending); NRAS / HRAS not explicitly reported (ngs_pending); Consensus molecular subtype (CMS) / CDX2 not reported (ngs_pending)
- **Efficacy/toxicity weight:** 0.7

13 ranked therapeutic options across 9 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Mcrc Backbone 1L interventions

1. Continue FOLFIRI + bevacizumab through the peri-operative window (bev hold 4-6 weeks pre-CRS and 4-6 weeks post-CRS); discuss FOLFOXIRI + bev intensification per TRIBE/TRIBE2 if oxaliplatin-naïve and ECOG 1 confirmed

Likelihood of effect: High — replicated phase 3 OS signal across AVF2107g, BICC-C, MAVERICC, TRIBE2; ESMO MCBS 3.

Toxicity burden: Moderate (G3+ neutropenia ~25-30%, G3+ diarrhea ~14% on FOLFIRI; cumulative neuropathy and bev wound-healing constraints with intensification)

Counter-productive MoA:Low (Bevacizumab wound-healing impairment in the peri-operative window — managed by the 4-6 week hold; not a mechanism-level blunting of the cytotoxic backbone.)

Overall:The replicated 1L OS backbone with the longest safety dossier in the case; the bev peri-op hold is the load-bearing safety thread, and FOLFOXIRI intensification is a permitted upgrade for a fit 37-year-old.**

**

Key references: PMID 15175435 · PMID 17947725 · PMID 25337750 · PMID 32007158 · PMID 30224341 · guideline:nccn-colon-v1.2026

Evidence in detail

Continue FOLFIRI + bevacizumab through the peri-operative window (bev hold 4-6 weeks pre-CRS and 4-6 weeks post-CRS); discuss FOLFOXIRI + bev intensification per TRIBE/TRIBE2 if oxaliplatin-naïve and ECOG 1 confirmed

****Hurwitz et al. *N Engl J Med* 2004****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

****Fuchs et al. *J Clin Oncol* 2007****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

****Loupakis et al. *N Engl J Med* 2014****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

****Cremolini et al. *Lancet Oncol* 2020****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

****Parikh et al. *Clin Cancer Res* 2019****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Pik3Ca M1043I interventions

1. Low-dose aspirin 160 mg PO QD as PIK3CA-pathway-altered adjunct (ALASCCA-derived), held 5-7 days pre-CRS and resumed post-op once surgical hemostasis allows

Likelihood of effect: Moderate — three concordant biomarker-stratified signals (ALASCCA HR 0.49, 80702 celecoxib OS HR 0.41, Liao 2012 HR 0.18) but indication gap from resected adjuvant to metastatic-equivalent

post-HIPEC.

Toxicity burden: Low (G3+ GI bleed ~3% vs 2% on ALASCCA; manageable with PPI + perioperative hold)

Counter-productive MoA:Low (Aspirin hold required around bev cycles and CRS-HIPEC to avoid additive bleeding; mechanism direction itself is supportive, not counter-productive.)

Overall:The only RoB2:Low phase 3 RCT in the dossier with a biomarker match — a cheap, layered, guideline-adopted adjunct that almost any other pick stacks on top of.****

Key references: PMID 40979555 · PMID 23094721 · PMID 38889377 · guideline:nccn-colon-v4.2025

2. Inavolisib 9 mg PO QD + bevacizumab on INTRINSIC (NCT04929223) PIK3CA-mutant arm — enroll at progression past FOLFIRI-bev or post-CRS-HIPEC, contingent on M1043I somatic + clonal confirmation from the rank-1 NGS

Likelihood of effect: Moderate ASSUMING M1043I confirmed somatic + clonal — INAVO120 PFS HR 0.43 is cross-tumor breast; CRC efficacy is hypothesis-grade (Juric 2018 2 PRs / 134 CRC subset; INTRINSIC arm-specific PFS unpublished).

Toxicity burden: Moderate (G3+ hyperglycemia 5.6%, any-grade 58.6%; G3+ stomatitis 5.6%; G3+ diarrhea 3.7%; discontinuation 6.5%)

Counter-productive MoA:Moderate (Critic and consensusite dissented on mechanism grounds: KRAS A59T co-activation may blunt single-pathway PI3K-alpha inhibition; cross-tumor breast extrapolation does not address CRC's downstream-RAS escape biology.)

Overall:The cleanest PIK3CA M1043I mechanism match in the dossier, gated by cross-tumor extrapolation and KRAS A59T co-activation — trial-route only, sequenced at progression rather than 1L displacement.****

Key references: PMID 39476340 · PMID 29401002 · PMID 31699932 · NCT04929223

Evidence in detail

Low-dose aspirin 160 mg PO QD as PIK3CA-pathway-altered adjunct (ALASCCA-derived), held 5-7 days pre-CRS and resumed post-op once surgical hemostasis allows

****Frödin et al. *N Engl J Med* 2025****

- **Disease context:** Resected stage I-III rectal or stage II-III colon cancer with somatic PI3K-pathway alteration (PIK3CA / PIK3R1 / PTEN) (adj) — n=626 randomized; 1,103 of 2,980 screened (37%) carried a PI3K-pathway alteration. Group A = PIK3CA exon 9/20 mut; Group B = other PI3K-pathway alterations.
- **n:** 626
- **Toxicities:** GI bleeding (grade ≥3): 3%; **Serious AEs overall** (grade any): 5%
- **Efficacy:** HR_RFS 0.49 (95% CI 0.24-0.98); HR_RFS 0.42 (95% CI 0.21-0.83); RFS 92.3% at 3 yr; RFS 85.9% at 3 yr

****Liao et al. *N Engl J Med* 2012****

- **Disease context:** Resected colorectal cancer stratified by tumor PIK3CA mutation (adj) — 964 stage I-IV CRC patients from NHS + HPFS; tumor PIK3CA mutation status assessed retrospectively; 17% PIK3CA-mut.
- **n:** 964
- **Toxicities:** Aspirin GI-bleed / ulcer risk class-typical; not formally captured in this cohort analysis.
- **Efficacy:** HR_OS 0.18 (95% CI 0.06-0.61); HR_OS 0.54 (95% CI 0.31-0.94); HR_OS 0.96 (95% CI 0.69-1.32)

****Yang et al. *J Clin Oncol* 2024****

- **Disease context:** Resected stage III colon cancer, PIK3CA-activating subset (adj) — Pre-specified biomarker subgroup of CALGB/SWOG 80702 (n=2,524 randomized total; ~16% PIK3CA-activating).
- **n:** 401
- **Toxicities:** Celecoxib G3+ AEs class-typical (cardiovascular, GI). Cohort safety dominated by FOLFOX backbone toxicity.
- **Efficacy:** HR_DFS 0.56 (95% CI 0.35-0.88); HR_OS 0.41 (95% CI 0.21-0.79); HR_DFS 1.07 (95% CI 0.89-1.28)

Inavolisib 9 mg PO QD + bevacizumab on INTRINSIC (NCT04929223) PIK3CA-mutant arm — enroll at progression past FOLFIRI-bev or post-CRS-HIPEC, contingent on M1043I somatic + clonal confirmation from the rank-1 NGS

****Jhaveri et al. *N Engl J Med* 2024****

- **Disease context:** HR+/HER2- PIK3CA-mutated metastatic breast cancer (1L after endocrine therapy) (1L) — n=325 with PIK3CA alterations confirmed by ctDNA or tissue.
- **n:** 325
- **Toxicities:** Hyperglycemia (grade ≥3): 5.6%; Hyperglycemia (grade any): 58.6%; Stomatitis (grade ≥3): 5.6%; Diarrhea (grade ≥3): 3.7%; Rash (grade any): 25%; Treatment discontinuation due to AE (grade any): 6.5%
- **Efficacy:** HR_PFS 0.43 (95% CI 0.32-0.59); PFS 15.0 months; PFS 7.3 months; ORR 58%; ORR 25%

****Juric et al. *J Clin Oncol* 2018****

- **Disease context:** Advanced solid tumors with PIK3CA alterations (mixed: breast, CRC, gynecologic, endometrial) (2L+) — n=134 PIK3CA-altered solid tumors; CRC subset included. PIK3CA mutations spanned the canonical helical / kinase hotspots (E542K, E545K, H1047R) and rarer alleles.
- **n:** 134
- **Toxicities:** Hyperglycemia (grade ≥3): 20%; Rash/maculopapular (grade ≥3): 10%; Diarrhea (grade ≥3): 7%; Nausea (grade any): 56%
- **Efficacy:** ORR 6%; RR 2 PRs

Peritoneal Loco Regional interventions

1. Cytoreductive surgery at a high-volume peritoneal-surface-malignancy center; if HIPEC is performed, mitomycin C HIPEC rather than oxaliplatin HIPEC, with PRODIGE 7 framing in the consent conversation

Likelihood of effect: Moderate for CRS (NCCN-endorsed selected-patient mOS 30-40 mo); Low for the HIPEC component specifically (PRODIGE 7 negative for oxaliplatin; no positive randomized signal for mitomycin C).

Toxicity burden: High (CRS-HIPEC 60-day morbidity ~25-40% G3+; bleeding, anastomotic leak, infection, prolonged ileus; treatment-related mortality 1-3% at high-volume centers, higher at lower-volume)

Counter-productive MoA: **Moderate** (Critic dissent stands on mechanism grounds: oxaliplatin HIPEC negative randomized OS readout; mitomycin C HIPEC has no positive randomized OS readout either — the procedure window itself delays systemic therapy.)

Overall: ****CRS retains the OS spine for resectable peritoneal CRC; HIPEC sits outside randomized OS validation — mitomycin C is the defensible perfusate at a high-volume center if HIPEC is elected.****

Key references: PMID 33417845 · PMID 31272834 · PMID 32717181 · PMID 39550351 · PMID 20689948 · NCT04861558 · NCT04779554 · guideline:nccn-colon-v1.2026

2. Oxaliplatin HIPEC framed as OS-positive — NOT RECOMMENDED per PRODIGE 7 (negative OS at HR ~1.00) reinforced by COLOPEC and PROPHYLOCHIP

Likelihood of effect: None over CRS alone — PRODIGE 7 negative for OS; COLOPEC and PROPHYLOCHIP reinforce the negative.

Toxicity burden: High (60-day G3+ morbidity 25-40%, treatment-related mortality 1-3%; without OS benefit)

Counter-productive MoA: **High** (Veto stands on mechanism: oxaliplatin HIPEC adds operative-window morbidity that delays systemic therapy without offsetting the patient with an OS gain.)

Overall: ****Closed on randomized evidence — surgical team retains autonomy on HIPEC agent choice; oxaliplatin perfusate specifically should not be framed as OS-validated in the family conversation.****

Key references: PMID 33417845 · PMID 31272834 · PMID 32717181

Evidence in detail

Cytoreductive surgery at a high-volume peritoneal-surface-malignancy center; if HIPEC is performed, mitomycin C HIPEC rather than oxaliplatin HIPEC, with PRODIGE 7 framing in the consent conversation

****Quenet et al. *Lancet Oncol* 2021****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

****Klaver et al. *Lancet Gastroenterol Hepatol* 2019****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

****Goéré et al. *Lancet Oncol* 2020****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

****Rovers et al. *Lancet Oncol* / ASO 2024****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Oxaliplatin HIPEC framed as OS-positive — NOT RECOMMENDED per PRODIGE 7 (negative OS at HR ~1.00) reinforced by COLOPEC and PROPHYLOCHIP

****Quenet et al. *Lancet Oncol* 2021****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

****Klaver et al. *Lancet Gastroenterol Hepatol* 2019****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

****Goéré et al. *Lancet Oncol* 2020****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Mss Peritoneal Immune interventions

1. Botensilimab + balstilimab at 2L+/3L+ after FOLFIRI-bev progression — operational paths: USC NCT06336902 (BOT/BAL + fasting-mimicking diet + IV vitamin C in KRAS-mutant MSS mCRC) is most actionable; Agenus EAP (NCT06751524) once standard lines exhausted; CCTG BATTMAN NCT07152821 is the registrational readout pending

Likelihood of effect: Moderate in the NLM peritoneal-dominant subset (ORR 22%, mOS 20.9 mo Bullock 2024) — strongest MSS-CRC ICI signal of the decade but selection bias + no BSC comparator until BATTMAN reads out.

Toxicity burden: Moderate (G3+ colitis 12%, G3+ hepatitis 8%, G3+ pyrexia in subset; cumulative G3+ irAE

35-38%)

Counter-productive MoA: Moderate (Critic dissented on evidence-quality mechanism (post-hoc NLM subgroup); conservative dissented on stacking research-only FMD + IV-C adjuncts on Fc-enhanced anti-CTLA-4 in active peritoneal disease.)

Overall: **Strongest modern MSS-CRC immunotherapy signal mapped to the patient's NLM peritoneal phenotype; 2L+ trial-route only, gated by FOLFIRI-bev progression and travel logistics.**

Key references: PMID 38871975 · NCT07152821 · NCT06336902 · NCT06751524

Evidence in detail

Botensilimab + balstilimab at 2L+/3L+ after FOLFIRI-bev progression — operational paths: USC NCT06336902 (BOT/BAL + fasting-mimicking diet + IV vitamin C in KRAS-mutant MSS mCRC) is most actionable; Agenus EAP (NCT06751524) once standard lines exhausted; CCTG BATMAN NCT07152821 is the registrational readout pending

****Bullock et al. *Nat Med* 2024****

- **Disease context:** Relapsed/refractory MSS metastatic colorectal cancer (2L+) — n=101 r/r MSS mCRC, mixed liver / non-liver metastatic distribution. Most patients had ≥3 prior lines.
- **n:** 101
- **Toxicities:** Diarrhea/colitis (grade ≥3): 12%; **Fatigue** (grade any): 45%; **Pyrexia** (grade any): 38%; **Hepatitis (transaminitis)** (grade ≥3): 8%; **Treatment-related AE ≥3** (grade ≥3): 35%; **Treatment-related death** (grade 5): 0% (n=101)
- **Efficacy:** ORR 17%; **ORR** 22%; **ORR** 5%; **OS** 20.9 months; **DoR** NR months

Her2 Amplification interventions

1. Trastuzumab deruxtecan (T-DXd) 5.4 mg/kg IV Q3W on the DESTINY-CRC02 pathway — CONDITIONAL on HER2 IHC 3+ or ISH ratio ≥2.0 returning positive from the rank-1 workup; RAS-mutant-allowed per the 2024 pan-tumor accelerated approval

Likelihood of effect: High in HER2 IHC 3+ subset (DESTINY-CRC02 ORR ~38% including RAS-mutant); contingent on HER2 status — foreclosed if HER2 negative (3-5% prior probability of positive).

Toxicity burden: Moderate (G3+ neutropenia ~30%, G3+ ILD 5-10% with treatment-related deaths reported, G3+ nausea/vomiting 10-15%)

Counter-productive MoA: Low (On-target ILD is the dominant counter-productive vector; RAS-mutant subset has lower absolute response than RAS-WT but A59T-specific behavior unreported.)

Overall: **The single most-actionable wildcard if HER2 amplification returns positive — RAS-mutant-allowed under the 2024 pan-tumor accelerated approval, and the cleanest HER2 path for an A59T patient.**

Key references: PMID 39058319 · PMID 27108243 · NCT04744831

Evidence in detail

Trastuzumab deruxtecan (T-DXd) 5.4 mg/kg IV Q3W on the DESTINY-CRC02 pathway — CONDITIONAL on HER2 IHC 3+ or ISH ratio ≥ 2.0 returning positive from the rank-1 workup; RAS-mutant-allowed per the 2024 pan-tumor accelerated approval

****Sartore-Bianchi et al. *Lancet Oncol* 2016****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Kras A59T interventions

1. Daraxonrasib (RMC-6236) on RASolve-GI (NCT06445062) — GATED on Revolution Medicines confirming KRAS A59T eligibility; JAB-23E73 (NCT06973564, Jacobio, KRAS-alteration-agnostic by public text) as the fallback inquiry; S241656 (NCT05786924, Servier ERK inhibitor, codon-agnostic) as the secondary fallback

Likelihood of effect: Low — zero published A59T clinical efficacy in any RAS(ON) or pan-RAS program; mechanism plausibility from Knox/Jiang Nature 2024 preclinical PDX work only.

Toxicity burden: Moderate (class-typical RAS(ON) acneiform rash 30-40% any-grade, GI nausea/diarrhea, hepatic; cumulative CRC profile maturing)

Counter-productive MoA:High (Critic veto stands on evidence-quality / mechanism grounds: ranking a sponsor-confirmation-gated n=0 A59T pick above replicated category-1 backbones inverts the evidence hierarchy.)

Overall:The asymmetric-upside single phone call for the atypical KRAS A59T allele — sponsor-confirmation-gated and critic-vetoed; document the inquiry result before any other RAS(ON) action.****

Key references: NCT06445062 · NCT06973564 · NCT05786924 · PMID 38778097 · PMID 38778099 · PMID 34031055

2. Cetuximab or panitumumab — default AVOID at 1L per Sorich 2015 meta-analysis; ASCEND-CRC platform (NCT07318389) research-framework rechallenge defensible only at later lines if a specific A59T functional argument is made and the rank-1 NGS confirms full RAS / BRAF clearance

Likelihood of effect: Low at 1L (Sorich 2015 OS HR 1.08 in any-new-RAS-mutant n=5,948); uncharacterized at later lines under a research framework with A59T-specific functional argument.

Toxicity burden: Moderate (G3+ acneiform rash 10-20%, hypomagnesemia, paronychia; manageable but real)

Counter-productive MoA:High (Conservative + consensus veto stands on mechanism: A59T is treated as

RAS-activating by NCCN consensus, so anti-EGFR is mechanism-counter-productive at the population level.)

Overall:Default AVOID at 1L per population-level meta-analysis; structured research-framework rechallenge defensible at later lines on Arena 2021 functional grounds — not welded shut, not 1L-recommendable.****

Key references: PMID 25115304 · PMID 24806288 · PMID 34031055 · PMID 30538852 · guideline:nccn-colon-v1.2026

3. Sotorasib (CodeBreaK 300) + panitumumab and adagrasib (KRYSTAL-1) + cetuximab — NOT RECOMMENDED for KRAS A59T (allele-incompatible)

Likelihood of effect: None — mechanism-incompatible at the covalent-warhead level; A59T lacks the switch-II cysteine the drugs target.

Toxicity burden: Moderate (class acneiform rash, GI nausea/diarrhea, hepatic) — but zero efficacy for A59T makes any AE counter-productive by definition.

Counter-productive MoA:High (Universal veto on mechanism grounds: drug binds G12C-specific cysteine; A59T lacks the warhead substrate entirely.)

Overall:Allele-incompatible — listed so the consultation team sees the closed door explicitly; the G12C-selective drugs cannot reach A59T at the covalent-warhead level.****

Key references: PMID 37870968 · PMID 36546659

Evidence in detail

Daraxonrasib (RMC-6236) on RASolve-GI (NCT06445062) — GATED on Revolution Medicines confirming KRAS A59T eligibility; JAB-23E73 (NCT06973564, Jacobio, KRAS-alteration-agnostic by public text) as the fallback inquiry; S241656 (NCT05786924, Servier ERK inhibitor, codon-agnostic) as the secondary fallback

****Arena et al. *Clin Cancer Res* 2021****

- **Disease context:**Metastatic colorectal cancer with atypical (non-G12/G13/Q61) KRAS or NRAS mutations (any) — Cohort of mCRC patients with atypical KRAS/NRAS alterations — A59, K117, A146, others — profiled functionally and clinically; subset received anti-EGFR therapy.
- **n:** 169
- **Toxicities:** Not assessed.
- **Efficacy:RR** intermediate descriptive

Cetuximab or panitumumab — default AVOID at 1L per Sorich 2015 meta-analysis; ASCEND-CRC platform (NCT07318389) research-framework rechallenge defensible only at later lines if a specific A59T functional argument is made and the rank-1 NGS confirms full RAS / BRAF clearance

****Sorich et al. *Ann Oncol* 2015****

- **Disease context:**Metastatic colorectal cancer stratified by extended-RAS status (KRAS exons 2/3/4 and NRAS exons 2/3/4) (any) — Pooled across CRYSTAL, PRIME, OPUS, FIRE-3, PEAK, 20020408, 20050181,

NORDIC-VII, MRC COIN — 5,948 patients; ~20% of KRAS exon-2 WT tumors carried a 'new RAS' alteration.

- **n:** 5948
- **Toxicities:** Class-typical anti-EGFR rash and hypomagnesemia plus chemo toxicity; safety was not the focus of this meta-analysis.
- **Efficacy:** **HR_OS** 0.87 (95% CI 0.77-0.99); **HR_PFS** 0.62 (95% CI 0.5-0.76); **HR_OS** 1.08 (95% CI 0.91-1.28); **HR_PFS** 1.03 (95% CI 0.78-1.36)

****Schirripa et al. *Int J Cancer* 2015****

- **Disease context:** **Metastatic colorectal cancer, chemo-refractory (2L+)** — n=786 mCRC patients; NRAS mutated in 47 (6%). NRAS-mut vs KRAS-mut groups compared for clinicopathology and anti-EGFR response.
- **n:** 786
- **Toxicities:** Retrospective; safety not assessed as a primary endpoint.
- **Efficacy:** **OS** 25.6 months; **ORR** 0%

****Arena et al. *Clin Cancer Res* 2021****

- **Disease context:** **Metastatic colorectal cancer with atypical (non-G12/G13/Q61) KRAS or NRAS mutations (any)** — Cohort of mCRC patients with atypical KRAS/NRAS alterations — A59, K117, A146, others — profiled functionally and clinically; subset received anti-EGFR therapy.
- **n:** 169
- **Toxicities:** Not assessed.
- **Efficacy:** **RR** intermediate descriptive

Sotorasib (CodeBreak 300) + panitumumab and adagrasib (KRYSTAL-1) + cetuximab — NOT RECOMMENDED for KRAS A59T (allele-incompatible)

****Fakih et al. *N Engl J Med* 2023****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

****Yaeger et al. *N Engl J Med* 2023****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Mcrc Backbone 2Lplus interventions

1. Post-CRS-HIPEC 2L+ bench: FOLFOX + bev (oxaliplatin-naïve lane preserved) → aflibercept-FOLFIRI (VELOUR) or ramucirumab-FOLFIRI (RAISE) anti-angiogenic switch → SUNLIGHT TAS-102 + bev → FRESCO-2 fruquintinib → regorafenib (CORRECT / CONCUR)

Likelihood of effect: Moderate-High — replicated phase 3 OS signal at each line; stacked sequencing 28-32 mo mOS achievable in fit patients.

Toxicity burden: Moderate (cumulative oxaliplatin neuropathy is rate-limiting; G3+ neutropenia and diarrhea at each line; anti-angiogenic AE compound)

Counter-productive MoA:Low (No mechanism-level counter-productive vector; cumulative neuropathy and anti-angiogenic compound AEs are patient-level toxicity, not therapeutic-goal blunting.)

Overall:The replicated 2L+ phase 3 OS floor the experimental picks have to beat — bake OPTIMOX in for the long young-patient runway and modulate by DPYD/UGT1A1 results.****

Key references: PMID 18421054 · PMID 25862517 · guideline:nccn-colon-v1.2026

Evidence in detail

Post-CRS-HIPEC 2L+ bench: FOLFOX + bev (oxaliplatin-naïve lane preserved) → aflibercept-FOLFIRI (VELOUR) or ramucirumab-FOLFIRI (RAISE) anti-angiogenic switch → SUNLIGHT TAS-102 + bev → FRESCO-2 fruquintinib → regorafenib (CORRECT / CONCUR)

Saltz et al. *J Clin Oncol* 2008

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Simkens et al. *Lancet* 2015

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Mss Tmb Low Phenotype interventions

1. Single-agent pembrolizumab, dostarlimab, nivolumab, or nivolumab + ipilimumab — NOT RECOMMENDED in MSS / TMB-low / PD-L1-negative mCRC

Likelihood of effect: None — phenotype-incompatible (MSS / TMB-low / PD-L1-neg); unless rank-1 MMR IHC unexpectedly returns dMMR.

Toxicity burden: Moderate (G3+ irAE colitis, hepatitis, pneumonitis, endocrinopathy) — counter-productive without target engagement.

Counter-productive MoA:High (Universal veto on phenotype grounds: ICI biology requires MSI-H or TMB-H or PD-L1-high substrate, none of which the patient presents.)

Overall:Phenotype-foreclosed in MSS / TMB-low / PD-L1-negative — listed so the door is explicitly closed; the rank-1 MMR IHC orthogonal is the only path that reopens it.****

Key references: PMID 26028255 · PMID 29355075 · guideline:nccn-colon-v1.2026

Evidence in detail

Single-agent pembrolizumab, dostarlimab, nivolumab, or nivolumab + ipilimumab — NOT RECOMMENDED in MSS / TMB-low / PD-L1-negative mCRC

****Le et al. *N Engl J Med* 2015****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

****Overman et al. *J Clin Oncol* 2018****

- **Disease context:** —
- **n:** —
- **Toxicities:** —
- **Efficacy:** —

Tp53 R273 interventions

1. Eprenetapopt (APR-246) and rezatapopt (PC14586) for TP53 R273 — NOT RECOMMENDED (eprenetapopt solid-tumor program closed; rezatapopt is Y220C-only by design)

Likelihood of effect: None — eprenetapopt solid-tumor program closed; rezatapopt is Y220C-restricted and KRAS-WT-only.

Toxicity burden: Low (none — drugs not accessible to this patient)

Counter-productive MoA:High (Universal veto on program-status + structural-design grounds: drug-allele mismatch at the binding-pocket level.)

Overall:Closed at the program + structural-design level; documented so the TP53 R273 reactivator question is explicitly addressed and the future ATR/WEE1 watch-list framing is preserved.****

Key references: PMID 40932470 · NCT04585750